

RSigELU: A nonlinear activation function for deep neural networks

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ABSTRACT

In deep learning models, the inputs to the network are processed using activation functions to generate the output corresponding to these inputs. Deep learning models are of particular importance in analyzing big data with numerous parameters and forecasting and are useful for image processing, natural language processing, object recognition, and financial forecasting. Sigmoid and tangent activation functions, which are traditional activation functions, are widely used in deep learning models. However, the sigmoid and tangent activation functions face the vanishing gradient problem. In order to overcome this problem, the ReLU activation function and its derivatives were proposed in the literature. However, there is a negative region problem in these activation functions. In this study, novel RSigELU activation functions, such as single-parameter RSigELU (RSigELUS) and double-parameter (RSigELUD), which are a combination of ReLU, sigmoid, and ELU activation functions, were proposed. The proposed RSigELUS and RSigELUD activation functions can overcome the vanishing gradient and negative region problems and can be effective in the positive, negative, and linear activation regions. Performance evaluation of the proposed RSigELU activation functions was performed on the MNIST, Fashion MNIST, CIFAR-10, and IMDB Movie benchmark datasets. Experimental evaluations showed that the proposed activation functions perform better than other activation functions.

1. Introduction

Deep learning models are widely used in application domains such as image processing, natural language processing, object recognition, translation, financial forecasting and the importance of these models is increasing every day (Schmidhuber, 2015). The deep learning algorithms that create models using training data are used to estimate the unknown outcomes using input data. Deep learning architectures consist of several layers including convolution, pooling, normalization, fully-connected, and activation layers.

Activation functions are used for healthy functioning of neural network architectures by considering the speed of the models during the training phase, the local minima, and the improved accuracy of the models. Activation functions form the basis for neural networks to model and learn the complex relationships between variables and layers. A well-designed activation function is effective on the performance of deep learning models. The activation layers process the inputs coming into the network and produce the output corresponding to these inputs,

using the activation functions. Activation functions should be appropriate for the data characteristics. There are numerous studies on the design of activation functions in the literature. Activation functions used in deep learning are expected to be non-linear and differentiable (Bircanoğlu & Arica, 2018). In deep learning studies, activation functions, such as ReLU and LReLU, are frequently used (Glorot et al., 2011; Krizhevsky et al., 2012; Maas et al., 2013) and in neural network studies, activation functions, such as sigmoid and tangent, are frequently used (Costarelli & Spigler, 2018; Costarelli, 2019; Costarelli & Vinti, 2017; Costarelli & Sambucini, 2018).

When an activation function is linear, it acts as linear regression with one-dimensional artificial neural network. However, since deep learning architectures are commonly associated with complex real-world problems, the linear activation functions fail to produce successful results. Therefore, nonlinear activation functions are preferred in multi-layered deep neural networks for learning significant features from the data (Bircanoğlu & Arica, 2018). In the deep neural network, the parameters are updated using the backpropagation algorithm. Since the derivatives

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are returned in the update process, the activation function used in the architecture must be differentiable. In addition, updates in neurons towards the lower layers become difficult in the deep neural network architectures used recently. This results in the vanishing gradient problem. This problem poses a major obstacle to deep learning in deep neural networks (Ebrahimi & Abadi, 2018). This obstacle raises the possibility of being stuck in local minima. This is because, as the network deepens, the vanishing gradient problem arises as frequently observed in the sigmoid and hyperbolic tangent activation functions. The difficulty in the training phase of the deep neural network can be eliminated by overcoming this problem. For this purpose, ReLU (Nair & Hinton, 2010), LReLU (Maas et al., 2013), PReLU (He et al., 2015), ELU (Clevert et al., 2015), SELU (Klambauer et al., 2017), Hexpo (Kong & Takatsuka, 2017), and LISA (Bawa & Kumar, 2019) were proposed in the literature to overcome the problem. However, these activation functions face the negative region problem. The ReLU activation function (Maas et al., 2013), introduced to overcome the vanishing gradient problem, sets negative inputs to zero when processing according to positive values. Thus, the learning process does not take place since the derivatives of negative outputs cannot be obtained. Values in the negative region are neglected during the learning process. To overcome this problem, studies such as LReLU (Maas et al., 2013), PReLU (He et al., 2015), and ELU (Clevert et al., 2015) were conducted.

In this study, new and novel RSigELU activation functions, such as single-parameter RSigELU (RSigELUS) and double-parameter RSigELU (RSigELUD), which are combinations of ReLU, sigmoid, and ELU activation functions, were proposed. Proposed RSigELUS and RSigELUD activation functions overcome the vanishing gradient and negative region problems to improve the learning process. The proposed RSigELU activation functions exhibit the behavior of a combination of ReLU and sigmoid functions in the positive activation region, while exhibiting the behavior of ELU function in the negative activation region and the linear function in the linear activation region (Bawa & Kumar, 2019). Experimental evaluations were conducted using the MNIST, Fashion MNIST, IMDB Movie, and CIFAR-10 benchmark datasets. Contributions of the study can be listed as follows.

1. New and novel RSigELU activation functions, such as RSigELUS and RSigELUD, which are combinations of ReLU, sigmoid, and ELU activation functions, were proposed.

2. The proposed RSigELU activation functions are capable of working in positive, negative, and linear activation regions, and overcome the vanishing gradient problem and negative region problem. The proposed RSigELU activation functions can work with single and double parameters, called RSigELUS and RSigELUD, respectively.

3. Performance evaluation of the proposed activation functions were carried out using a VGG architecture on MNIST, Fashion MNIST, IMDB Movie, and CIFAR-10 benchmark datasets.

In the second part of this study a literature review is presented. The problem definition is given in the third section and the proposed activation functions are introduced in the fourth section. The approaches used to evaluate the activation functions and datasets are discussed in the fifth section. Experimental results are presented in the sixth section and the results obtained are discussed in the conclusions section.

2. Literature REVIEW

Studies that focus on the design of activation functions for deep learning structures have focused on the vanishing gradient and negative region problems. In this study, activation functions are proposed for the solution of both problems.

The ReLU (Nair & Hinton, 2010) activation function is able to deal with the vanishing gradient problem. ReLU overcomes the vanishing gradient problem by returning a positive value when a negative value comes to the input, without making any changes (Eq. (1)). The drawback of the ReLU activation function, however, is that it returns zero for the values in the negative region. In addition, it also causes some neurons to

become useless since it is not zero-oriented. To overcome the negative value problem in ReLU, the Leaky ReLU (Maas et al., 2013), SELU (Klambauer et al., 2017), ELU (Clevert et al., 2015), PReLU (He et al., 2015), DReLU (Hendrycks & Gimpel, 2016a, 2016b), etc. activation functions were proposed. The Leaky ReLU activation function (Eq. (2)) proposed by Maas et al. (2013) multiplies the negative value in the input region of the data by a constant alpha value (such as 0.2) to allow its contribution to the learning process and also overcomes the problem related to the ReLU and sigmoid functions. The problems of negative value and vanishing gradient were overcome by using trainable parameters based on the data instead of using constant alpha values in other activation functions, such as PReLU (Eq. (3)) and Leaky ReLU (He et al., 2015). The ELU activation function (Eq. (4)) proposed by Clevert et al. (2015) aims to overcome both bias and variance problems and to overcome negative value and vanishing gradient problems by using an exponential structure. The PELU activation function (Eq. (5)) developed by Trotter et al. (2017) was developed with the aim of controlling the exponential structure in the ELU function. The SELU activation function (Eq. (6)) was proposed by Klambauer et al. (2017) to overcome the vanishing gradient, negative value, and poor initial value of ELU. The SELU activation function overcomes the above-mentioned problems by adding a constant scale value for this purpose, unlike the ELU activation function. Godin et al. (2018) proposed the DReLU activation function (Eq. (7)) to train deeper models. Similar to the ELU activation function, the DReLU activation function can operate in both positive and negative regions, exhibits the properties of both ReLU and hyperbolic tangent functions, and can overcome the vanishing gradient problem. In their study, Qiumei et al. (2019) proposed the FELU activation function (Eq. (8)) with the aim of increasing the classification accuracy and computational speeds of the ELU and ReLU activation functions. The proposed FELU activation function is able to overcome the vanishing gradient problem and allows the network to run faster using an exponential function to accelerate computation in the negative region. Ramachandran et al. (2017) the Swish activation function was proposed as an alternative to the ReLU function to accommodate the sigmoid function in order to overcome the vanishing gradient problem. Of the parameters shown in Equation 1–8, x is input data, α is scale constant learning parameter, and b is degradation and α is saturation parameters in Eqs. (5) and (8), where $\alpha, b > 0$.

$$f(x) = \begin{cases} 0, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases} \quad (1)$$

$$f(x) = \begin{cases} 0.01x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases} \quad (2)$$

$$f(x) = \begin{cases} \alpha x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases} \quad (3)$$

$$f(x) = \begin{cases} \alpha(e^x - 1), & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases} \quad (4)$$

$$f(x) = \begin{cases} \alpha \left(e^{\frac{x}{b}} - 1 \right), & \text{if } x < 0 \\ \frac{a}{b}x, & \text{if } x \geq 0 \end{cases} \quad \alpha, b > 0 \quad (5)$$

$$f(x) = \text{Scale} \begin{cases} ae^x - a, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases} \quad (6)$$

$$f(x, y) = \begin{cases} 0, & \text{if } x \leq 0 \text{ and } y \leq 0 \\ x, & \text{if } x > 0 \text{ and } y \leq 0 \\ -y, & \text{if } x \leq 0 \text{ and } y > 0 \\ x - y, & \text{if } x > 0 \text{ and } y > 0 \end{cases} \quad (7)$$

$$f(x) = \begin{cases} x, & \text{if } x > 0 \\ \alpha \left(2^{\frac{x}{\alpha}} - 1 \right), & \text{if } x \leq 0 \end{cases} \quad (8)$$

Wang et al. (2019) proposed the ReLTanh activation function (Eq. (9)) proposed to solve the vanishing gradient problem provides learning in both regions by replacing the positive and negative regions of the hyperbolic tangent activation function with each other and taking their derivatives. Kong et al. (2017) proposed the Hexpo activation function (Eq. (10)) was proposed to solve the vanishing gradient problem. It focused on the solution of the activation function vanishing gradient problem proposed by Bawa and Kumar (2019) (Eq. (11)) and Sharma (2019) (Eq. (12)). Liu et al. (2016) proposed the Softplus activation function (Eq. (13)) was proposed as an alternative to the sigmoid and hyperbolic tangent activation functions. Glorot and Bengio (2010) proposed the softsign activation function (Eq. (14)) was proposed as an alternative to the hyperbolic tangent activation function for easy differentiation compared to the sigmoid and tangent functions. Of the parameters in Equation 9–14, x is input data, α_1 and α_2 are the constant learning parameters, $\tanh'$ is tangent hyperbolic first derivative, and α , b , c and $d > 0$ constant values are used to overcome the vanishing gradient problem.

$$f(x) = \begin{cases} \tanh'(\lambda^+)(x - \lambda^+) + \tanh(\lambda^+), & x \geq \lambda^+ \\ \tanh(x), & \lambda^- < x < \lambda^+ \\ \tanh'(\lambda^-)(x - \lambda^-) + \tanh(\lambda^-), & x \leq \lambda^- \end{cases} \quad (9)$$

$$f(x) = \begin{cases} -\alpha \left(e^{\frac{x}{\alpha}} - 1 \right), & \text{if } x \geq 0 \\ c \left(e^{\frac{x}{c}} - 1 \right), & \text{otherwise} \end{cases} \quad (10)$$

$$f(x) = \begin{cases} \alpha_1 x - \alpha_1 + 1, & \text{if } 1 < x < \infty \\ x, & \text{if } 0 \leq x \leq 1 \\ \alpha_2 x, & \text{if } -\infty < x < 0 \end{cases} \quad (11)$$

$$f(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ \frac{x}{1 + e^{-x}}, & \text{if } x > 0 \end{cases} \quad (12)$$

$$f(x) = \ln(1 + e^x) \quad (13)$$

$$f(x) = \frac{x}{(1 + |x|)} \quad (14)$$

The activation functions proposed in the literature address the vanishing gradient problem and the negative region. In our study, we propose new RSigELUS ve RSigELUD activation functions to overcome vanishing gradient and negative region problems and with the aim of increasing the classification accuracy of the ELU and ReLU activation functions.

3. Problem statement

In this study, novel RSigELU activation functions were proposed to overcome the vanishing gradient and negative region problem that has become the focus of attention in deep learning models. In the deep neural networks, the networks are barely updated or become useless due to the present problems. This process prevents the network from deepening and learning, and even may cause to use too much computational power for training. Thus, this may cause a fall into local minima in deep neural networks (Wang et al., 2019). The vanishing gradient problem was discovered by Hochreiter (1998). This problem occurs when a large number of derivatives are taken in each process in cascaded deep learning models. The gradient disappears as the derivatives shrink as the model begins to deepen, causing a decline in the learning process. Thus, the vanishing gradient problem is observed in the model due to the sigmoid and tangent activation functions commonly used in deep

networks. The sigmoid function produces an output between 0 and 1, as in Fig. 1. This function produces meaningful expressions in the interval of $[-1, 1]$, while in its next operation the results converge to 0 and it stops learning. In addition, the same problem is also true for the hyperbolic tangent function. As shown in Fig. 2, after the values in the interval of $[-1, 1]$, like in the sigmoid function, the result of the derivative operation converges to 0 (Bengio et al., 1994). In order to overcome this problem, it must be replaced with differentiable functions. Thus, it backpropagates the errors by providing a continuous learning process. There are numerous studies in the literature to overcome the vanishing gradient problem (Maas et al., 2013; Klambauer et al., 2017; Clevert et al., 2015; He et al., 2015; Hendrycks & Gimpel, 2016a, 2016b; Wang et al., 2019; Kong et al., 2017; Bawa & Kumar, 2019; Sharma, 2019). The behaviors of these activation functions in these studies are shown in Fig. 3.

Nair and Hinton (2010) proposed the ReLU activation function, introduced to overcome the vanishing gradient problem, sets negative inputs to zero when processing according to positive values. Thus, the learning process does not take place since derivatives of the negative outputs cannot be obtained. In order to overcome the problem mentioned in this study and to incorporate negative values into the process, new activation function of RSigELUS and RSigELUD were proposed. The proposed RSigELU activation functions are discussed in the next section.

4. Proposed activation functions

This study proposes new activation functions, RSigELUS and RSigELUD, which can operate in accordance with the data in three active regions (i.e. positive, linear, and negative) in order to overcome the vanishing gradient and negative region problems. The proposed RSigELUS activation function requires a single parameter and RSigELUD activation function requires double parameters. In order to capture the characteristics of both regions, single- and double-parameter structures were used in the study. Of these single and double parameters, the parameters of α and β refers to the slope coefficients. With these slope coefficients, the positive region is controlled by α , while the negative region is controlled with β . The behavior of the proposed RSigELUS and RSigELUD activation functions are shown in Fig. 4.

4.1. RSigELUS activation function

The single-parameter RSigELU activation function (RSigELUS) is defined in Eq. (15).

$$f(x) = \begin{cases} x \left(\frac{1}{1 + e^{-x}} \right) \alpha + x, & \text{if } 1 < x < \infty \\ x, & \text{if } 0 \leq x \leq 1 \\ \alpha(e^x - 1), & \text{if } -\infty < x < 0 \end{cases} \quad (15)$$

The RSigELUS function in the Eq. (15) shows three different active

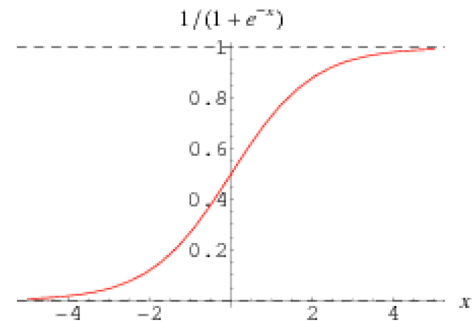


Fig. 1. Sigmoid function.

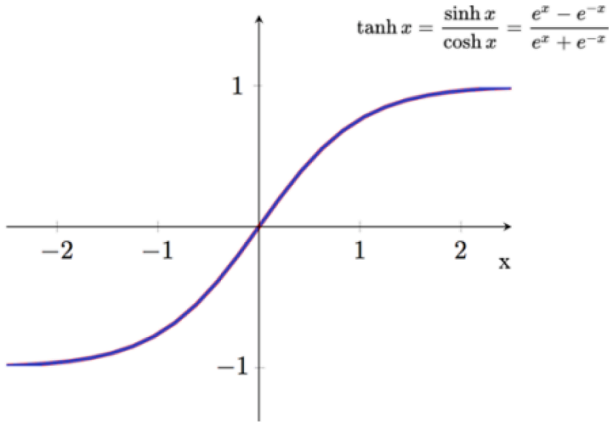


Fig. 2. Hyperbolic tangent function.

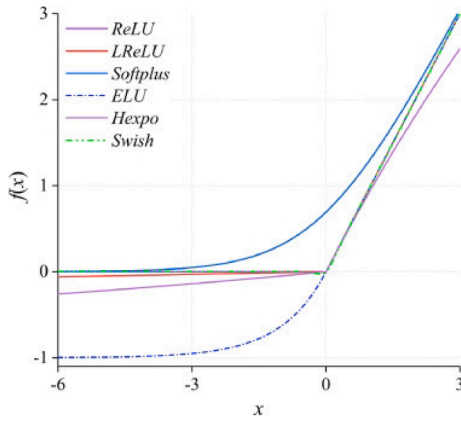


Fig. 3. Behaviors of some activation functions in the literature (Wang et al., 2019).

regions. All other training parameters of the proposed RSigELUS activation function were kept constant and the experiment was performed by changing the α slope variable between $0 < \alpha < 1$. This value is used to control the positive and negative regions. Here, x represents the input to the activation function. The linear activity region maps input to output, directly. The proposed RSigELUS activation function behaves as shown in Fig. 4-a. The sigmoid function for the positive region was used to have a continuous learning state in deep models through a hybrid structure

with the ReLU function. It is used as a linear activation function for the linear region. It also acts as the ELU activation function for the negative region. The proposed single-parameter RSigELUS activation function can exhibit different behaviors according to the values given. When the parameter a becomes 0 ($\alpha = 0$) in the RSigELUS activation function, the function exhibits ReLU behavior, as in the Eq. (16).

$$f(x) = \begin{cases} x, & \text{if } 1 < x < \infty \\ x, & \text{if } 0 \leq x \leq 1 \\ 0, & \text{if } -\infty < x < 0 \end{cases} \quad (16)$$

The derivative of the proposed RSigELUS activation function is shown in the Eq. (17).

$$\frac{df(x)}{dx} = \begin{cases} \frac{-\alpha x}{(e^x + 1)^2} + \frac{\alpha x - \alpha}{(e^x + 1)} + \alpha + 1, & \text{if } 1 < x < \infty \\ 1, & \text{if } 0 \leq x \leq 1 \\ \alpha e^x, & \text{if } -\infty < x < 0 \end{cases} \quad (17)$$

4.2. RSigELUD activation function

The proposed double-parameter RSigELUD activation function is shown in the Eq. (18).

$$f(x) = \begin{cases} x \left(\frac{1}{1 + e^{-x}} \right) \alpha + x, & \text{if } 1 < x < \infty \\ x, & \text{if } 0 \leq x \leq 1 \\ \beta(e^x - 1), & \text{if } -\infty < x < 0 \end{cases} \quad (18)$$

Three different active regions are shown in Eq. (18). The linear activity region maps input to output directly. In the study, β slope coefficient can take values in the range of $0 < \beta < 1$ and α slope coefficient can take values in the range of $0 < \alpha < 1$. This value is used to control the positive and negative regions. The best scale values were presented in Tables 2, 3, 4, and 5. According to the results obtained in experimental studies, the value of β was found as $\beta = 0.2$ and used in all experiments. The derivative of the proposed RSigELUD activation function is shown in the Eq. (20). As shown in Fig. 4 for both functions, the learning process is continuous since both proposed activation functions are differentiable. The activation function proposed in Eq. (18) can act as a different activation function according to the given values. For example, for the values $\alpha = 0$ and $\beta = 0$, the system acts as the ReLU activation function as in Eq. (16). In addition, when $\alpha = 0$ and $\beta = 0.2$, the function acts as if the ELU activation function, as in Eq. (19).

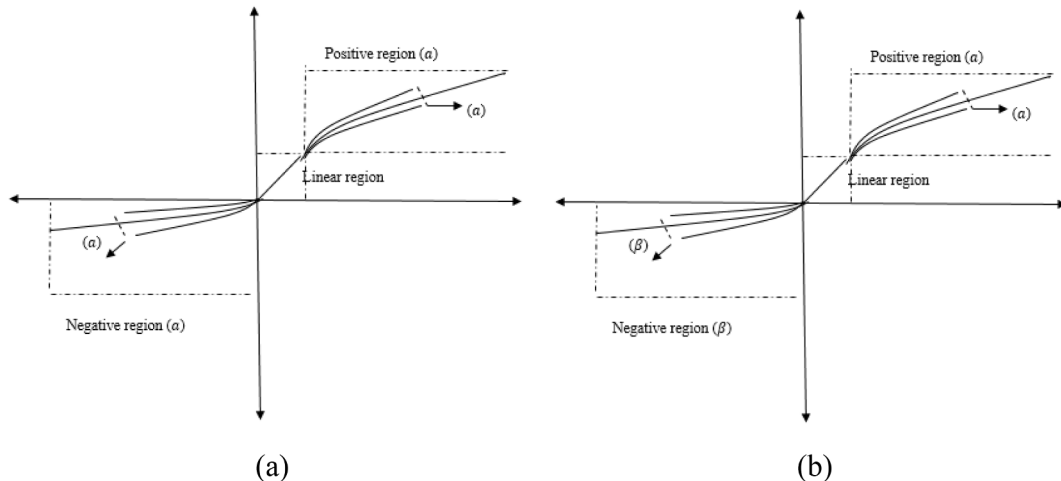


Fig. 4. Behavior of the proposed (a) RSigELUS and (b) RSigELUD activation functions.

Table 1
Parameters of the VGG architecture used in the study.

CNN Layers	Parameters		
Input Data			
Conv1	32@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
Conv1	32@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
BatchNormalization			
MaxPooling1	Pool size = 2	Stride = 2	
Dropout	Rate 0.4%		
Conv2	48@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
Conv2	48@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
BatchNormalization			
MaxPooling2	Pool size = 2	Stride = 2	
Dropout	Rate 0.4%		
Conv3	64@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
Conv3	64@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
Conv3	64@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
BatchNormalization			
MaxPooling3	Pool size = 2	Stride = 2	
Dropout	Rate 0.4%		
Conv4	96@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
Conv4	96@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
Conv4	96@3 × 3 filter	Padding = same	regularizers.l2 = 0.0005
BatchNormalization			
MaxPooling3	Pool size = 2	Stride = 2	
Dropout	Rate 0.4%		
Flatten			
Fully connected	512		
Fully connected	512		
Softmax			

$$f(x) = \begin{cases} x, & \text{if } 0 \leq x \leq \infty \\ \beta(e^x - 1), & \text{if } -\infty < x < 0 \end{cases} \quad (19)$$

The derivative of the proposed RSigELUD activation function is shown in the Eq. (20).

$$\frac{df(x)}{dx} = \begin{cases} \frac{-\alpha x}{(e^x + 1)^2} + \frac{\alpha x - \alpha}{(e^x + 1)} + \alpha + 1, & \text{if } 1 < x < \infty \\ 1, & \text{if } 0 \leq x \leq 1 \\ \beta e^x, & \text{if } -\infty < x < 0 \end{cases} \quad (20)$$

In deep neural networks, it is important to have differentiable activation functions since the backpropagation architecture is used as the basis for updating parameters during training. When the proposed single- and double-parameter activation functions are examined, it can be stated that the activation functions are active in both the positive and negative regions after the derivative operation. The linear activity region maps input to output, directly.

5. Material and method

The study was carried out using Tensorflow Backend and Keras Library and Google Colab platform to evaluate the classification performances of deep learning algorithms using the activation functions proposed in this study and in the literature. The study was conducted using a system with the Intel Core i5 7200U 2.5 GHz processor, and 12 GB DDR3 memory.

In the following part of this section, the benchmark datasets used in the study were introduced, information was given about the convolutional neural network used in the study, and finally, experimental results were presented and the results were discussed.

5.1. Benchmark datasets

In this study, experimental evaluations were performed on MNIST, Fashion MNIST, CIFAR-10, and IMDB Movie benchmark datasets.

The MNIST dataset consists of a total 70,000 handwritten digits and 10 classes (LeCun et al., 1998). In this study, 60,000 of the images contained in the dataset were used for training and 10,000 of the images were used for testing. Each image in the dataset is grayscale and 28 × 28 pixels in size. No augmentation was performed to the MNIST data in the experimental evaluation.

Table 2
Experimental results of RSigELUD for MNIST dataset.

Alpha(α)	Scale (β)									
	0.1		0.2		0.3		0.4		0.5	
	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.
0.1	0.0537	0.9935	0.0465	0.9946	0.0449	0.9945	0.0495	0.9935	0.0651	0.9927
0.2	0.0562	0.9929	0.0390	0.9946	0.0529	0.9941	0.0540	0.9935	0.0596	0.9944
0.3	0.0467	0.9935	0.0356	0.9948	0.0410	0.9931	0.0550	0.9931	0.0489	0.9939
0.4	0.0427	0.9941	0.0360	0.9945	0.0548	0.9936	0.0510	0.9939	0.0548	0.9934
0.5	0.0457	0.9940	0.0458	0.9947	0.0498	0.9938	0.0616	0.9926	0.0491	0.9937

Table 3
Experimental results of RSigELUD for Fashion-MNIST dataset.

Alpha(α)	Scale (β)									
	0.1		0.2		0.3		0.4		0.5	
	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.
0.1	0.5812	0.9250	0.337	0.9252	0.5353	0.9135	0.5558	0.9167	0.5733	0.9190
0.2	0.6023	0.9243	0.4985	0.9256	0.5335	0.9220	0.5265	0.9216	0.5204	0.9190
0.3	0.5748	0.9221	0.6519	0.9257	0.5522	0.9144	0.4610	0.9151	0.4732	0.9205
0.4	0.5898	0.9222	0.4755	0.9255	0.5440	0.9223	0.4468	0.9222	0.4091	0.9209
0.5	0.5566	0.9255	0.4700	0.9244	0.4888	0.9162	0.4296	0.9216	0.5174	0.9155

Table 4

Experimental results of RSigELUD for CIFAR-10 dataset.

Alpha(α)	Scale (β)									
	0.1		0.2		0.3		0.4		0.5	
	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.
0.1	0.6307	0.8353	0.6787	0.8358	0.6524	0.8284	0.6214	0.8327	0.6658	0.8325
0.2	0.6416	0.8323	0.6663	0.8346	0.6837	0.8179	0.6647	0.8343	0.6750	0.8238
0.3	0.6129	0.8318	0.6897	0.8325	0.6124	0.8310	0.6508	0.8291	0.7277	0.8016
0.4	0.6995	0.8317	0.6746	0.8331	0.6466	0.8300	0.6481	0.8326	0.6655	0.8310
0.5	0.7029	0.8161	0.6095	0.8421	0.6405	0.8316	0.6083	0.8377	0.6548	0.8215

Table 5

Experimental results of RSigELUD for IMDb Movie dataset.

Alpha(α)	Scale (β)									
	0.1		0.2		0.3		0.4		0.5	
	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.	Val. Loss	Val. Acc.
0.4	0.7906	0.8101	0.8604	0.8411	0.7011	0.8356	0.7247	0.8411	0.7525	0.8422
0.5	0.8547	0.8387	0.8077	0.8407	0.7615	0.8404	0.7208	0.8400	0.7150	0.8327
0.6	0.8920	0.8320	0.7333	0.8330	0.9370	0.8221	0.7603	0.8284	0.7388	0.8208
0.7	0.7920	0.8339	0.6230	0.8357	0.7372	0.8330	0.7617	0.8264	0.7489	0.8353
0.8	0.6197	0.8354	0.7337	0.8372	0.7573	0.8370	0.7540	0.8311	0.7217	0.8326
0.9	0.7038	0.8296	0.6614	0.8392	0.7499	0.8305	0.6997	0.8338	0.7433	0.8266

The Fashion MNIST dataset, developed by the Zalando research team, consists of 70,000 clothing images, as an alternative to the MNIST dataset (Xiao et al., 2017). In this study, 60,000 of the images contained in the dataset were used for training and 10,000 of the images were used for testing. The dataset has 10 classes, and each image is grayscale and 28×28 pixels in size.

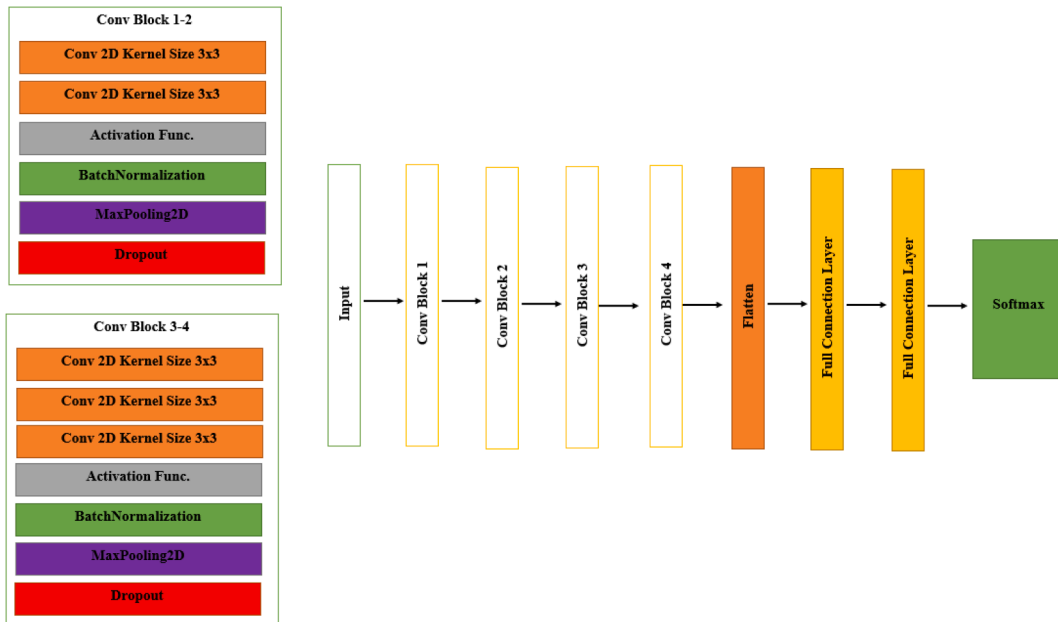
The CIFAR-10 dataset consists of 60,000 color images in total (Krizhevsky & Hinton, 2009). The dataset consists of 10 classes, and each image is 32×32 pixels in size. In this study, 50,000 of the images were used for training and 10,000 of the images were used for testing.

The IMDb Movie dataset consists of 25,000 records about movies. In this study, 25,000 of the IMDb Movie dataset were used as training data (12,500 positive and 12,500 negative tagged) and 25,000 as test data (12,500 positive and 12,500 negative tagged).

5.2. Convolutional neural network

The Convolutional Neural Network (CNN), developed by LeCun et al. in 1990, is a multi-layer neural network model inspired by the human brain (LeCun et al., 1990; LeCun et al., 1990, 2015). In this architecture, the convolution expression combines two signals mathematically, producing the third signal. We used the VGG-based CNN architecture to evaluate the proposed RSigELU activation functions. Basically, CNN architecture consists of convolution layer, activation layer, pooling layer, normalization layer, dropout layer, flatten layer, fully connected layer, and the softmax layer. The VGG architecture used is shown in Fig. 5.

In Fig. 5, the six-layer convolutional block consists of a flatten layer, two fully connected layers, and a softmax layer. It consists of a 3×3 filter in the convolution blocks. In order to keep the process cost

**Fig. 5.** VGG-based CNN architecture for evaluating the proposed activation function.

minimum, filter size is chosen as 3×3 . It utilizes the proposed RSigELU activation functions to deal with the vanishing gradient and negative region problems. Following the convolution process, the normalization layer was used to normalize the dataset. This was followed by maximum pooling to reduce the size of the input data in the next stage. It also simplifies the output of nonlinear data and reduces the number of parameters that must be learned by the network. The maximum pooling was applied with the values $\text{pool_size} = 2$ and $\text{stride} = 2$. The stride value performs one or more bitwise shifting on the data converted to the matrix structure during the convolution process. This process directly affects the output size. After the pooling process, the dropout layer was used to prevent memorization in the model and its value was set to 0.4. Dropout is a regularization technique that prevents neural networks from over-fitting. The flatten layer is used to prepare the data for the fully connected layer. In order to connect it to the previous neurons through the fully-connected layer, the cascaded neuron density was set to 512 neurons. The fully connected layer takes the data from the flattening process and performs the learning process via the neural network. Finally, the softmax layer was preferred to enhance the classification success (Kiliçarslan et al., 2020; LeCun et al., 1990, 1998; Murphy, 2012). The softmax function outputs the probability of each class that should be among the classes with the highest probability from all classes. Classification is done by the softmax layer placed at the end of the architecture. Softmax is an extended version of the logistic regression and a probability-based linear classifier for two or more class problems. In the convolution neural network, updating the weights correctly is important for the learning process. To update the weights, a back-propagation algorithm was used. In the study, Adadelta was used as an optimizer.

Table 1 shows 32 kernels in the first convolution block in the architecture used in this study. The second convolution block consists of 48 kernels, the third convolution block consists of 64 kernels, and the last block consists of 96 kernels. In all experiments, VGG architecture was kept constant.

6. Experimental results and discussion

The study presents the experimental evaluations of the proposed RSigELU activation functions on MNIST, Fashion MNIST, CIFAR-10, and IMDB Movie datasets. In the double parameters (RSigELUD) case, the best results of the experiments performed on each dataset to determine the scale value are shown in Tables 2–5. The best success rates of the proposed RSigELUS and RSigELUD activation functions and other known activation functions for the MNIST dataset are presented in Tables 6–8. In the tables best train loss (Train Loss), train accuracy (Train Acc.), validation loss (Val. Loss), and validation accuracy (Val. Acc.) values are given. Figs. 6–9 show the average validation loss (Val. Loss) values of the activation function results and their standard

Table 7

The best VGG success rates of RSigELUD for MNIST dataset.

Alpha (α)	Scale (β)	Epoch	Train Loss	Train Acc	Val Loss	Val acc	Time (s)
0.05	0.2	40	0.0038	0.9993	0.0322	0.9953	327
0.1	0.2	40	0.0022	0.9996	0.0465	0.9946	364
0.15	0.2	40	0.0024	0.9996	0.0504	0.9951	360
0.2	0.2	40	0.0024	0.9995	0.0390	0.9945	362
0.25	0.2	40	0.0021	0.9996	0.0320	0.9951	363
0.3	0.2	40	0.0024	0.9995	0.0356	0.9948	362
0.35	0.2	40	0.0028	0.9993	0.0416	0.9947	362
0.4	0.2	40	0.0034	0.9994	0.0360	0.9945	362
0.45	0.2	40	0.0034	0.9994	0.0353	0.9944	363
0.5	0.2	40	0.0023	0.9995	0.0458	0.9947	366

deviations for each dataset. In the experiments, the parameters values of activation functions used for comparison were set at the ones given in the study conducted by Bawa and Kumar (2019).

According to the results of the experiments in Tables 2–5, it was observed that the best success rates were obtained at the scale value of $\beta = 0.2$ in all RSigELUD experiments. Therefore, the scale value in the study was set to $\beta = 0.2$.

Fig. 6 shows the average validation loss values of the activation function results and their standard deviations for the MNIST dataset. As can be seen in the figure, the average validation loss value of the proposed RSigELUS activation function is the smallest and it has the smallest variability. The average validation loss value of the proposed RSigELUD activation function is smaller than that of other activation functions except that of LISA, Softplus, Softsing. As can be seen in Fig. 6, the proposed RSigELUS ve RSigELUD activation functions yielded results consistent the ELU and ReLU activation functions and their variations of LReLU, SELU, and GELU for the MNIST dataset.

Tables 6–8 show that the best validation accuracy (Val. Acc.) and validation loss (Val. Loss) values were 0.9957 and 0.0254, respectively, according to the experimental results on the MNIST dataset for 40 epochs and $\alpha = 0.2$ parameter value in the RSigELUD activation function. The values were obtained after repeating the experiments for 5 times. Moreover, the validation accuracy and validation loss values were found to be 0.9953 and 0.0322, respectively, for the proposed RSigELUD activation function for the parameter values of 0.05 and 0.2. As can be seen in the table, the proposed RSigELUS and RSigELUD activation functions outperform other activation functions and RSigELUS outperforms RSigELUD in terms of validation accuracy. In experiments performed with LReLU and SELU activation functions in the literature, the validation accuracy value was found as 0.9951. However, our proposed activation functions seem to work more slowly than the activation functions known in the literature.

The best success rates of the proposed RSigELUS and RSigELUD activation functions and other known activation functions for the Fashion MNIST dataset are presented in Tables 9–11.

Fig. 7 shows the average validation loss values of the activation function results and their standard deviations for the Fashion MNIST dataset. As can be seen in the figure, the average validation loss value of the proposed RSigELUS activation function is smaller than that of ReLU, LReLU, Swish, ELU, SELU, GELU, Hexpo, Softplus, and the proposed RSigELUD and bigger than that of Softsing and LISA. The average validation loss value of the proposed RSigELUD activation function is smaller than that of other activation functions except that of ReLU, LReLU, Softsing, SELU, LISA, Hexpo, and the proposed RSigELUS. As can be seen in Fig. 7, the proposed RSigELUS yielded results consistent the ReLU and ELU activation functions and their variations of LReLU, SELU, and GELU for the Fashion MNIST dataset.

As shown in Tables 9–11, the proposed RSigELUS activation function has the best success rate for the Fashion MNIST dataset. The best success rate was obtained with the parameter value of 0.05 in the RSigELUS activation function, and the validation accuracy and validation loss

Table 6

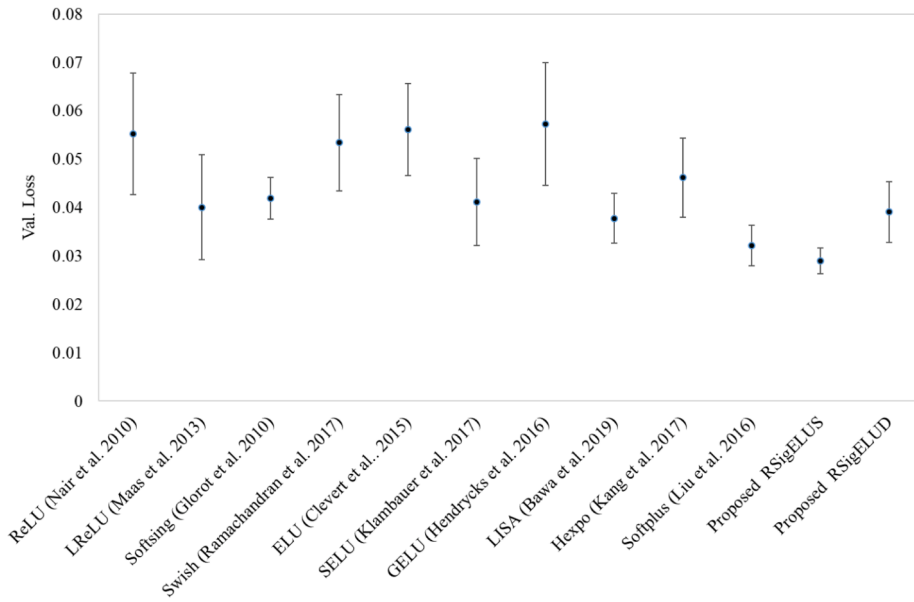
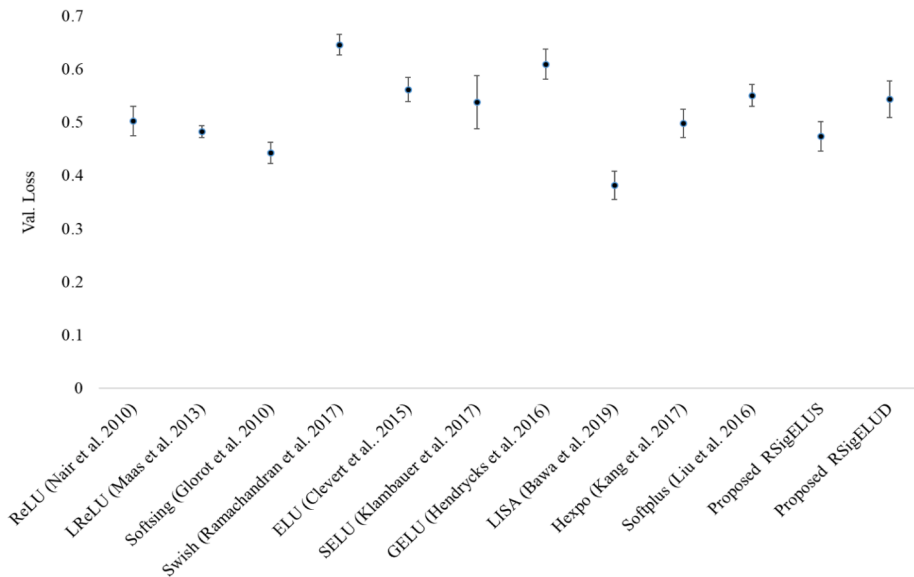
The best VGG success rates of RSigELUS for MNIST dataset.

Alpha (α)	Epoch	Train Loss	Train Acc.	Val. Loss	Val. Acc.	Time (s)
0.01	40	0.0032	0.9990	0.0284	0.9949	290
0.02	40	0.0026	0.9995	0.0270	0.9953	321
0.03	40	0.0023	0.9994	0.0282	0.9955	325
0.04	40	0.0016	0.9995	0.0264	0.9948	320
0.05	40	0.0026	0.9994	0.0302	0.9944	325
0.1	40	0.0029	0.9991	0.0306	0.9955	480
0.15	40	0.0030	0.9992	0.0297	0.9942	442
0.2	40	0.0033	0.9990	0.0254	0.9957	290
0.25	40	0.0037	0.9989	0.0236	0.9947	661
0.3	40	0.0043	0.9990	0.0326	0.9949	662
0.35	40	0.0052	0.9986	0.0378	0.9938	683
0.4	40	0.0050	0.9987	0.0298	0.9940	334
0.45	40	0.0060	0.9985	0.0306	0.9941	323
0.5	40	0.0081	0.9981	0.0425	0.9936	324

Table 8

The best VGG success rates of activation functions for MNIST dataset.

Activation	α	β	Epoch	Train Loss	Train Acc.	Val. Loss	Val. Acc.	Time (s)
ReLU (Nair et al., 2010)			40	0.0019	0.9995	0.0326	0.9946	200
LReLU (Maas et al., 2013)	0.2		40	0.0027	0.9994	0.0277	0.9951	201
Softsing (Glorot et al., 2010)			40	0.0055	0.9985	0.0417	0.9933	201
Swish (Ramachandran et al., 2017)			40	0.0024	0.9993	0.0340	0.9946	200
ELU (Clevert et al., 2015)	0.2		40	0.0133	0.9970	0.0614	0.9917	205
SELU (Klambauer et al., 2017)			40	0.0037	0.9990	0.0290	0.9951	284
GELU (Hendrycks et al., 2016)			40	0.0010	0.9996	0.0396	0.9945	401
LISA (Bawa et al., 2019)	0.15		40	0.0010	0.9998	0.0321	0.9945	324
Hexpo (Kong et al., 2017)			40	0.0008	0.9999	0.0354	0.9949	326
Softplus (Liu et al., 2016)			40	0.0075	0.9974	0.0288	0.9936	201
Proposed RSigELUS	0.2		40	0.0033	0.9990	0.0254	0.9957	290
Proposed RSigELUD	0.05	0.2	40	0.0038	0.9993	0.0322	0.9953	327

**Fig. 6.** Average validation loss (Val. Loss) values of activation function results and their standard deviations for MNIST dataset.**Fig. 7.** Average validation loss (Val. Loss) values of activation function results and their standard deviations for Fashion MNIST dataset.

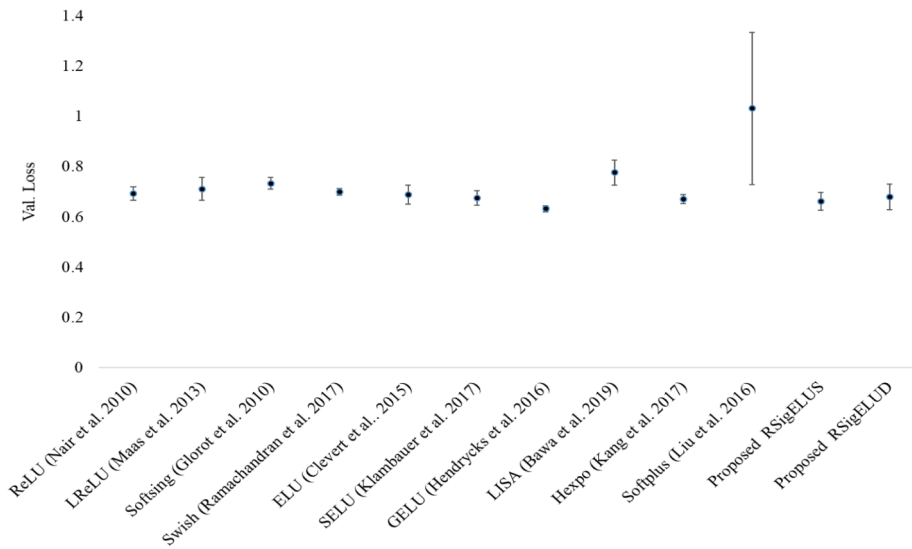


Fig. 8. Average validation loss (Val. Loss) values of activation function results and their standard deviations for CIFAR-10 dataset.

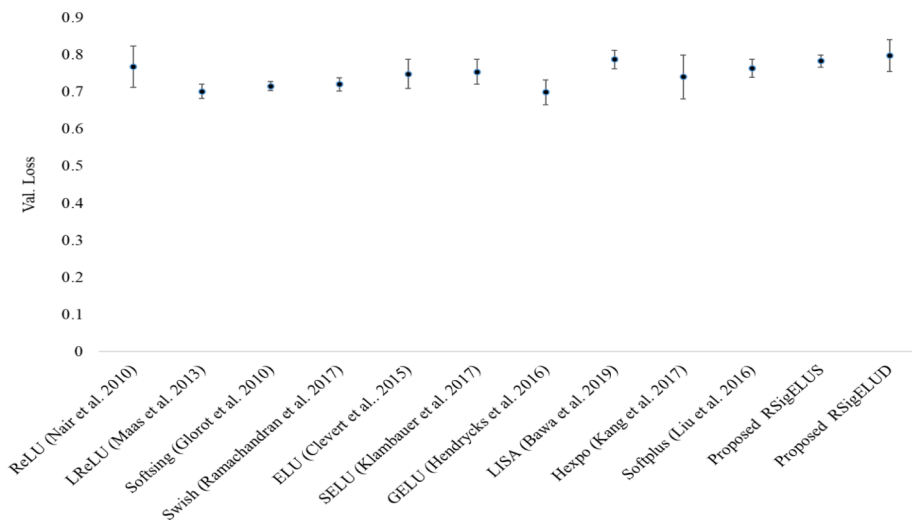


Fig. 9. Average validation loss (Val. Loss) values of activation function results and their standard deviations for IMDB Movie dataset.

values were found as 0.9277 and 0.4682, respectively. As can be seen in the table, the proposed RSigELUS activation function outperforms other activation functions in terms of validation accuracy. The proposed RSigELUD activation function outperforms other activation functions, except LReLU and the proposed RSigELUS, in terms of validation accuracy. The proposed RSigELUD activation function was found to have a lower success rate for Fashion MNIST data compared to the LReLU activation function. These values were 0.9257 for the double-parameter structure, and 0.9266 for LReLU.

The success rates of the proposed RSigELUS and RSigELUD activation functions and other known activation functions for the CIFAR-10 dataset are presented in Tables 12–14.

Fig. 8 shows the average validation loss values of the activation function results and their standard deviations for the CIFAR-10 dataset. As can be seen in the figure, the average validation loss value of the proposed RSigELUS activation function is smaller than that of other functions except that of GELU. The average validation loss value of the proposed RSigELUD activation function is smaller than that of other activation functions except that of SELU, GELU, Hexpo, and RSigELUS. As can be seen in Fig. 7, the proposed RSigELUS ve RSigELUD activation functions yielded results consistent the ELU and ReLU activation

functions and ReLU variation of LReLU for the CIFAR-10 dataset.

Tables 12–14 show the best success and loss rates for the proposed RSigELUS and RSigELUD activation functions and other known activation functions for the CIFAR-10 dataset. According to the results, the best validation accuracy and validation loss values in the CIFAR-10 data set are 0.8430 and 0.5477, respectively, for 0.05 and 0.2 values in the proposed RSigELUD activation function. Moreover, the validation accuracy and validation loss values for the parameter value of 0.45 in the proposed RSigELUS activation function were found to be 0.8412 and 0.5920, respectively. As can be seen in the table, the proposed RSigELUS and RSigELUD activation functions outperform other activation functions, and RSigELUD outperforms RSigELUS in terms of validation accuracy. Other than these, the best success rate appears to be obtained by SELU, among the other activation functions. The best success rates of the proposed RSigELUS and RSigELUD activation functions and other known activation functions for the IMDB Movie dataset are presented in Tables 15–17.

Fig. 9 shows the average validation loss values of the activation function results and their standard deviations for the IMDB Movie dataset. As can be seen in the figure, the average validation loss value of

Table 9

The best VGG success rates of RSigELUS for Fashion MNIST dataset.

Alpha (α)	Epoch	Train Loss	Train Acc.	Val. Loss	Val. Acc.	Time (s)
0.01	40	0.0203	0.9932	0.3504	0.9273	369
0.02	40	0.0200	0.9946	0.5267	0.9260	367
0.03	40	0.0216	0.9936	0.4473	0.9257	368
0.04	40	0.0203	0.9939	0.5279	0.9266	369
0.05	40	0.0224	0.9933	0.4682	0.9277	365
0.1	40	0.0215	0.9938	0.5071	0.9254	370
0.15	40	0.0268	0.9926	0.4005	0.9276	372
0.2	40	0.0293	0.9917	0.5266	0.9259	380
0.25	40	0.0312	0.9904	0.6026	0.9235	374
0.3	40	0.0474	0.9848	0.4272	0.9239	374
0.35	40	0.0420	0.9879	0.4668	0.9239	361
0.4	40	0.0381	0.9884	0.5437	0.9217	362
0.45	40	0.0620	0.9799	0.4346	0.9202	362
0.5	40	0.0505	0.9836	0.4971	0.9220	363
0.001	40	0.0222	0.9931	0.5565	0.9273	362
0.002	40	0.0195	0.9938	0.4250	0.9254	363
0.003	40	0.0199	0.9938	0.5175	0.9244	364
0.004	40	0.0208	0.9936	0.5188	0.9236	364
0.005	40	0.0203	0.9938	0.4510	0.9269	367
0.006	40	0.0218	0.9933	0.5115	0.9242	366
0.007	40	0.0216	0.9934	0.4615	0.9264	369
0.008	40	0.0210	0.9937	0.4994	0.9275	370
0.009	40	0.0252	0.9925	0.5157	0.9237	370

Table 10

The best VGG success rates of RSigELUD for Fashion MNIST dataset.

Alpha (α)	Scale (β)	Epoch	Train Loss	Train Acc.	Val Loss	Val acc	Time (s)
0.05	0.2	40	0.0261	0.9926	0.5712	0.9232	372
0.1	0.2	40	0.0300	0.9912	0.337	0.9252	370
0.15	0.2	40	0.0254	0.9930	0.5543	0.9174	370
0.2	0.2	40	0.0275	0.9925	0.4985	0.9256	369
0.25	0.2	40	0.0291	0.9914	0.4694	0.9240	368
0.3	0.2	40	0.0277	0.9917	0.6519	0.9257	368
0.35	0.2	40	0.0273	0.9921	0.562	0.9257	367
0.4	0.2	40	0.0309	0.9911	0.4755	0.9255	366
0.45	0.2	40	0.0343	0.9896	0.5535	0.9234	365
0.5	0.2	40	0.0318	0.9910	0.4700	0.9244	366

the proposed RSigELUS activation function is smaller than that of LISA and the proposed RSigELUD and bigger than that of rest of the activation functions. The average validation loss value of the proposed RSigELUD activation function is bigger than that of other activation functions. As can be seen in Fig. 8, the proposed RSigELUS ve RSigELUD activation functions yielded results consistent the ELU and ReLU activation functions and ReLU variation of LReLU for the IMDB Movie dataset.

Tables 15–17 show the best success and loss rates for the proposed RSigELUS and RSigELUD activation functions and other known activation functions for the IMDB Movie dataset. According to the results, the best validation accuracy and validation loss values in the IMDB Movie

Table 11

The best VGG success rates of activation functions for Fashion MNIST dataset.

Activation	α	β	Epoch	Train Loss	Train Acc.	Val. Loss	Val. Acc.	Time (s)
ReLU (Nair et al., 2010)			40	0.0206	0.9938	0.4987	0.9243	285
LReLU (Maas et al., 2013)	0.2		40	0.0298	0.9913	0.4731	0.9266	288
Softsign (Glorot et al., 2010)			40	0.0276	0.9906	0.4500	0.9190	246
Swish (Ramachandran et al., 2017)			40	0.0248	0.9924	0.6259	0.9123	286
ELU (Clevert et al., 2015)	0.2		40	0.0201	0.9937	0.5620	0.9231	329
SELU (Klambauer et al., 2017)	0.1		40	0.0226	0.9930	0.5647	0.9139	292
GELU (Hendrycks et al., 2016)			40	0.0228	0.9932	0.5843	0.9159	411
LISA (Bawa et al., 2019)	0.15		40	0.0379	0.9871	0.4024	0.9234	369
Hexpo (Kong et al., 2017)			40	0.0238	0.9920	0.4892	0.9178	369
Proposed RSigELUS	0.05		40	0.0224	0.9933	0.4682	0.9277	365
Proposed RSigELUD	0.35	0.2	40	0.0273	0.9921	0.5620	0.9257	367

data set were 0.8448 and 0.7903, respectively, for the parameter value of 0.75 in the proposed RSigELUS activation function. Moreover, the validation accuracy and validation loss values for the parameter values of 0.4 and 0.2 in the proposed RSigELUD activation function were found to be 0.8411 and 0.8604, respectively. As can be seen in the table, the proposed RSigELUS and RSigELUD activation functions outperform other activation functions, and RSigELUS outperforms RSigELUD in terms of validation accuracy. Other than these, the best success rate appears to be obtained by GELU, among the other activation functions. The best validation accuracy and validation loss values in GELU were 0.8408 and 0.7032.

In our study, we proposed new RSigELUS and RSigELUD activation functions to overcome vanishing gradient and negative region problems and to increase the classification accuracy of ELU and ReLU activation functions. The proposed RSigELUS and RSigELUD activation functions were shown to yield the best results in experimental evaluations on MNIST, Fashion MNIST, CIFAR-10, and IMDB Movie datasets. Each datasets have different characteristics and, because of these reason, as can be seen in Figs. 6–9, none of the methods have consistent validation loss values for all the datasets. However, the proposed activation functions outperform most of the previously proposed activation functions

Table 12

The best VGG success rates of RSigELUS for CIFAR-10 dataset.

Alpha (α)	Epoch	Train Loss	Train Acc.	Val. Loss.	Val. Acc.	Time (s)
0.05	40	0.3811	0.9081	0.6772	0.8382	720
0.1	40	0.3839	0.9080	0.6610	0.8365	788
0.15	40	0.4156	0.8979	0.6876	0.8385	880
0.2	40	0.3799	0.9093	0.6764	0.8398	840
0.25	40	0.3807	0.9092	0.6825	0.8372	843
0.3	40	0.3861	0.9056	0.7212	0.8269	875
0.35	40	0.3895	0.9057	0.6866	0.8346	844
0.4	40	0.3968	0.9012	0.6783	0.8307	641
0.45	40	0.5030	0.8608	0.5920	0.8412	588
0.5	40	0.4926	0.8634	0.6256	0.8312	566

Table 13

The best VGG success rates of RSigELUD for CIFAR-10 dataset.

Alpha (α)	Scale (β)	Epoch	Train Loss	Train Acc	Val Loss	Val acc	Time (s)
0.05	0.2	40	0.3177	0.9021	0.5477	0.8430	728
0.1	0.2	40	0.3983	0.9030	0.6787	0.8358	625
0.15	0.2	40	0.3884	0.9059	0.6804	0.8345	620
0.2	0.2	40	0.4034	0.9014	0.6663	0.8346	609
0.25	0.2	40	0.3906	0.9043	0.6980	0.8294	648
0.3	0.2	40	0.3998	0.9012	0.6897	0.8325	566
0.35	0.2	40	0.4016	0.9003	0.6922	0.8283	572
0.4	0.2	40	0.3896	0.9031	0.6746	0.8331	569
0.45	0.2	40	0.3904	0.9018	0.6851	0.8297	648
0.5	0.2	40	0.4811	0.8678	0.6095	0.8421	644

Table 14

The best VGG success rates of activation functions for CIFAR-10 dataset.

Activation	α	β	Epoch	Train Loss	Train Acc	Val Loss	Val acc	Time (s)
ReLU (Nair et al., 2010)			40	0.5444	0.8511	0.6858	0.8161	640
LReLU (Maas et al., 2013)	0.2		40	0.3782	0.8799	0.5970	0.8202	655
Softsign (Glorot et al., 2010)			40	0.6264	0.8236	0.7167	0.8036	488
Swish (Ramachandran et al., 2017)			40	0.5108	0.8660	0.6701	0.8281	485
ELU (Clevert et al., 2015)	0.2		40	0.5455	0.8512	0.6409	0.8292	482
SELU (Klambauer et al., 2017)			40	0.4870	0.8658	0.6014	0.8387	523
GELU (Hendrycks et al., 2016)			40	0.4870	0.8640	0.6113	0.8342	532
LISA (Bawa et al., 2019)	0.15		40	0.6097	0.8240	0.6332	0.8218	562
Hexpo (Kong et al., 2017)			40	0.5765	0.8411	0.6364	0.8307	602
Softplus (Liu et al., 2016)			40	0.5640	0.8455	1.3497	0.6113	628
Proposed RSigELUS	0.45		40	0.5030	0.8608	0.5920	0.8412	588
Proposed RSigELUD	0.05	0.2	40	0.3177	0.9021	0.5477	0.8430	728

Table 15

The best VGG success rates of RSigELUS for IMDB Movie dataset.

Alpha (α)	Epoch	Train Loss	Train Acc.	Val. Loss	Val. Acc.	Time (s)
0.4	40	0.0300	0.9902	0.7540	0.8338	925
0.45	40	0.0068	0.9976	0.9519	0.8334	521
0.5	40	0.0095	0.9972	0.8737	0.8251	522
0.55	40	0.0178	0.9940	0.8103	0.8357	877
0.6	40	0.0210	0.9929	0.7885	0.8252	878
0.65	40	0.0212	0.9926	0.7764	0.8378	880
0.7	40	0.0195	0.9929	0.7394	0.8427	831
0.75	40	0.0204	0.9932	0.7903	0.8448	820
0.8	40	0.0372	0.9865	0.9033	0.8174	822
0.85	40	0.0332	0.9883	0.7173	0.8372	923
0.9	40	0.0104	0.9966	0.8002	0.8332	520

Table 16

The best VGG success rates of RSigELUD for IMDB Movie dataset.

Alpha (α)	Scale (β)	Epoch	Train Loss	Train Acc	Val Loss	Val acc	Time (s)
0.4	0.2	40	0.0215	0.9925	0.8604	0.8411	885
0.45	0.2	40	0.0241	0.9918	0.7855	0.8375	892
0.5	0.2	40	0.0202	0.9928	0.8077	0.8407	908
0.55	0.2	40	0.0206	0.9933	0.7287	0.8409	918
0.6	0.2	40	0.0218	0.9924	0.7333	0.8330	932
0.65	0.2	40	0.0255	0.9913	0.8020	0.8381	928
0.7	0.2	40	0.0245	0.9918	0.6230	0.8357	925
0.75	0.2	40	0.0211	0.9920	0.7702	0.8394	918
0.8	0.2	40	0.0231	0.9920	0.7337	0.8372	921
0.85	0.2	40	0.0215	0.9922	0.7778	0.8358	932
0.9	0.2	40	0.0215	0.9925	0.6614	0.8392	882

for the dataset of MNIST, Fashion MNIST, and CIFAR-10, in terms of validation losses. For IMDB dataset, the previously proposed activation functions outperform the proposed activation functions. When Fig. 9 is inspected for IMDB dataset, it can be seen that validation losses of all the

Table 17

The best VGG success rates of activation functions for IMDB Movie dataset.

Activation	α	β	Epoch	Train Loss	Train Acc	Val Loss	Val acc	Time (s)
ReLU (Nair et al., 2010)			40	0.0249	0.9915	0.7651	0.8324	895
LReLU (Maas et al., 2013)	0.2		40	0.0234	0.9928	0.6792	0.8322	926
Softsign (Glorot et al., 2010)			40	0.0221	0.9922	0.7312	0.8352	1148
Swish (Ramachandran et al., 2017)			40	0.0183	0.9940	0.7099	0.8356	1152
ELU (Clevert et al., 2015)	0.2		40	0.0235	0.9919	0.7464	0.8382	1147
SELU (Klambauer et al., 2017)	0.1		40	0.0223	0.9923	0.7727	0.8305	1134
GELU (Hendrycks et al., 2016)			40	0.0245	0.9916	0.7032	0.8408	1118
LISA (Bawa et al., 2019)	0.15		40	0.0216	0.9926	0.7588	0.8377	1142
Hexpo (Kong et al., 2017)			40	0.0218	0.9926	0.7453	0.8370	936
Softplus (Liu et al., 2016)			40	0.0271	0.9914	0.7989	0.8203	1169
Proposed RSigELUS	0.75		40	0.0204	0.9932	0.7903	0.8448	820
Proposed RSigELUD	0.4	0.2	40	0.0215	0.9925	0.8604	0.8411	885

activation functions are close to each other. In addition, as can be seen in Tables 8, 11, 14 and 17, the proposed activation functions outperform previously proposed activation functions in terms of the best validation accuracies.

7. Conclusions

The activation layers process the inputs coming into the network and produce the output corresponding to these inputs, using the activation functions. Since deep learning architectures commonly concern with complex problems, linear activation functions are not appropriate for such architectures. This is because, activation functions form the basis for the neural networks to learn and predict the complex and continuous relationship between variables. Therefore, nonlinear activation functions come to the fore in deep learning models. However, there is the problem of vanishing gradient in the widely known sigmoid and tangent activation functions. In order to overcome this problem, ReLU activation function was proposed. However, the ReLU activation function cannot return a correct value for the negative region since it outputs zero for the negative values. To overcome the negative value problem in ReLU, alternative activation functions have been proposed. Hence, we proposed new single- and double-parameter RSigELU activations (such as, RSigELUS and RSigELUD, respectively) to overcome both the vanishing gradient and negative value problems in our study. The proposed activation functions comes to the fore with their features of increasing the success rate in the training process. Experiments were carried out on the MNIST, Fashion MNIST, CIFAR-10, and IMDB Movie datasets with the proposed RSigELU activation functions, and they appears to produce better results than that of in the literature.

In the future work, we plan to use intelligent optimization methods (Celik et al., 2016; Koylu et al., 2013; Ozcan & Celik, 2018; Erkan, 2020) to obtain hyperparameters of deep learning algorithms and to apply proposed activation functions on other network structures using different datasets.

CRediT authorship contribution statement

Serhat Kiliçarslan: Conceptualization, Software, Visualization, Supervision, Writing - review & editing. **Mete Celik:** Conceptualization, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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